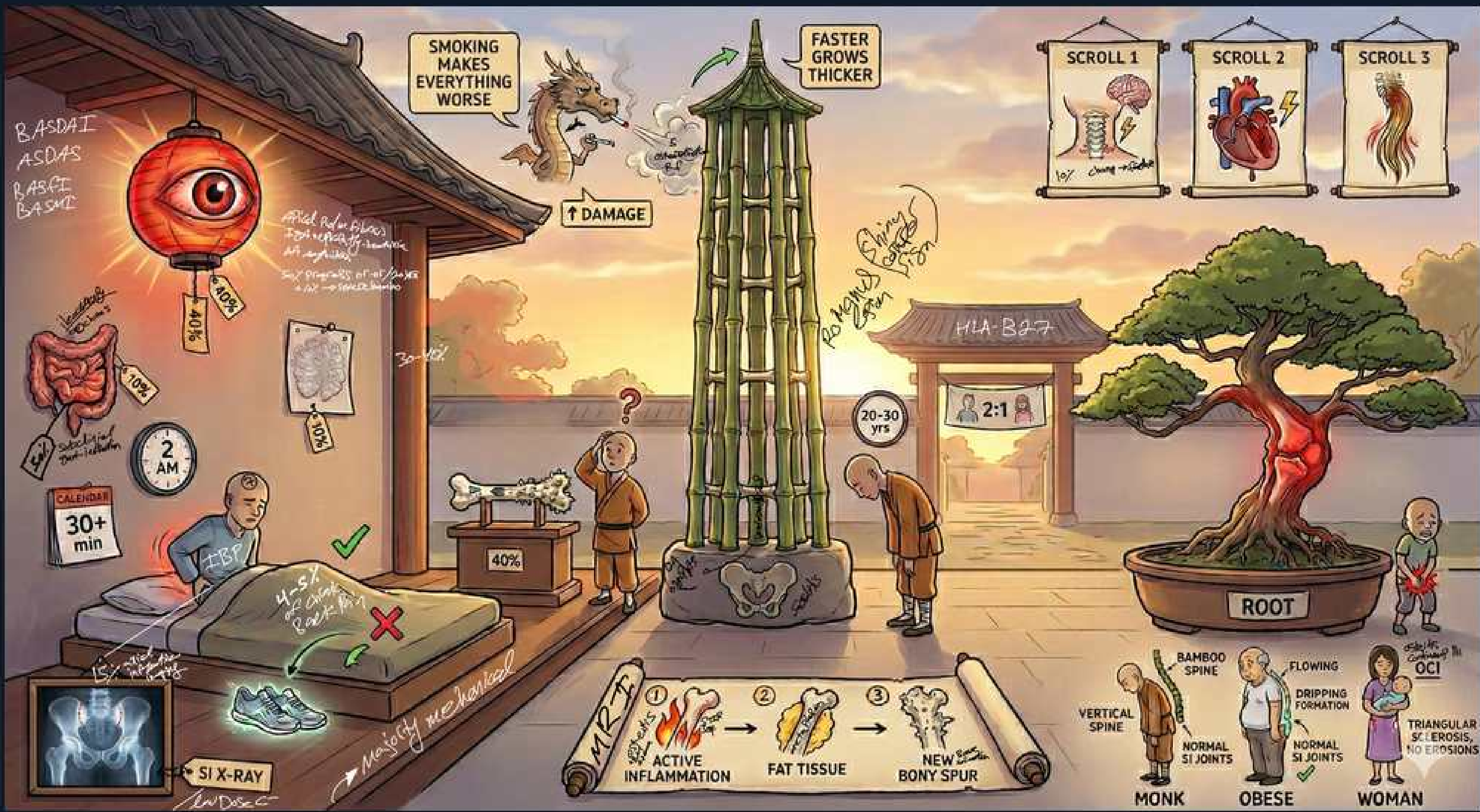
Spondyloarthritis — Axial SpA: Clinical Features, Imaging, Diagnosis, Complications



1. Red eye lantern

WHAT YOU SEE

Glowing red eye on a hanging lantern

WHAT IT ENCODES

Acute anterior uveitis (iritis) is the most common extra-articular manifestation of axial SpA, affecting up to 30–40% over a lifetime and strongly linked to HLA-B27. It is typically UNILATERAL, recurrent, and alternating, presenting with a painful red eye, photophobia, blurred vision, and a constricted/irregular pupil. Treat urgently with topical glucocorticoids and a cycloplegic; refer to ophthalmology to prevent synechiae.

BOARD TRAP

It is ANTERIOR uveitis (unilateral, acute, painful), not the bilateral/insidious posterior uveitis of sarcoidosis or the scleritis of rheumatoid arthritis/granulomatosis with polyangiitis. A painful red eye with photophobia in a young man with back pain is anterior uveitis — choosing conjunctivitis (which is painless without photophobia) is the trap.

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2. Inflamed bowel

WHAT YOU SEE

Inflamed red loops of intestine drawn at the left wall — the subclinical gut inflammation linking axial SpA to IBD and steering drug choice toward anti-TNF antibodies.

WHAT IT ENCODES

Subclinical gut inflammation is found on ileocolonoscopy in a majority of axial SpA patients, and ~5–10% develop overt IBD (Crohn or ulcerative colitis), part of the seronegative spondyloarthritis family. This shapes drug choice: anti-TNF MONOCLONAL ANTIBODIES (infliximab, adalimumab) treat both joints and gut, and IL-17 inhibitors should be avoided when IBD coexists.

BOARD TRAP

IL-17 inhibitors (secukinumab, ixekizumab) can WORSEN or trigger IBD and are contraindicated in active IBD — choosing an IL-17 blocker for an axial SpA patient with Crohn disease is the trap; use a monoclonal anti-TNF instead. Also note etanercept (a soluble receptor) is ineffective for IBD and uveitis, unlike the anti-TNF monoclonals.

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3. X-ray of pelvis

WHAT YOU SEE

A pelvic radiograph tag labeled SI X-RAY — the first imaging step; definite sacroiliitis here defines radiographic axial SpA (but a normal film never excludes it).

WHAT IT ENCODES

Plain pelvic radiograph of the sacroiliac joints is the first imaging step and defines RADIOGRAPHIC axial SpA (ankylosing spondylitis) when it shows definite sacroiliitis — erosions, sclerosis, joint-space pseudo-widening then narrowing, and eventual ankylosis (modified New York criteria). Structural change takes years to appear, so early disease is often radiographically silent.

BOARD TRAP

A NORMAL sacroiliac X-ray does NOT exclude axial SpA — early/non-radiographic disease requires MRI to detect bone-marrow edema (osteitis). Stopping the workup at a normal plain film and excluding the diagnosis is the classic imaging trap.

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4. 30+ min stiffness

WHAT YOU SEE

Clock tag reading 30+ min beside man in bed

WHAT IT ENCODES

Inflammatory back pain (the cardinal symptom) is defined by: insidious onset before age 45, duration >3 months, morning stiffness >30 minutes, improvement WITH EXERCISE but NOT with rest, pain worse at night/early morning (often waking the patient in the second half of the night), and good response to NSAIDs. Four of five features (ASAS criteria) strongly suggest inflammatory back pain.

BOARD TRAP

Mechanical/degenerative back pain is the wrong-answer foil: it WORSENS with activity, IMPROVES with rest, lacks prolonged (>30 min) morning stiffness, and does not wake the patient at night. The discriminator is 'better with exercise, worse with rest' — the reverse pattern points away from SpA.

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5. Smoking dragon

WHAT YOU SEE

Dragon blowing smoke labeled SMOKING MAKES EVERYTHING WORSE / DAMAGE

WHAT IT ENCODES

Smoking is a major modifiable risk factor: it accelerates radiographic spinal progression (more syndesmophytes), worsens function and disease activity, and blunts response to TNF inhibitors. Smoking cessation is a core management recommendation alongside physical therapy and exercise.

BOARD TRAP

Smoking accelerates structural DAMAGE and reduces treatment response in axial SpA — assuming it is irrelevant to a rheumatologic (non-pulmonary) disease is the trap. Counseling cessation is part of the correct management answer.

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6. Active inflammation flame

WHAT YOU SEE

Scroll step 1: flame labeled ACTIVE INFLAMMATION

WHAT IT ENCODES

Step 1 of the bone-remodeling cascade is active inflammation/osteitis at the entheses and subchondral bone, seen as BONE-MARROW EDEMA on STIR/T2 MRI — the earliest and REVERSIBLE stage. This is the imaging target in non-radiographic axial SpA and the rationale for early biologic therapy to halt progression to new bone.

BOARD TRAP

Bone-marrow edema on MRI is the early, treatable stage — it is NOT yet the irreversible bony fusion. Equating early MRI inflammation with end-stage bamboo spine, or assuming inflammation directly becomes bone without the fatty-metaplasia intermediate, is the trap.

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7. Bamboo pagoda

WHAT YOU SEE

Tall bamboo tower/pagoda growing in center, FASTER GROWS THICKER

WHAT IT ENCODES

'Bamboo spine' is the end-stage radiographic appearance of ankylosing spondylitis: vertically bridging marginal SYNDESMOPHYTES fuse adjacent vertebral bodies into a rigid, smooth column, often with squaring of vertebrae and fusion of facet joints and the ossified anterior longitudinal ligament. It signifies advanced disease with reduced spinal mobility (positive Schober test) and fracture risk.

BOARD TRAP

Bamboo-spine syndesmophytes are THIN, VERTICAL, and MARGINAL — distinguish them from the thick, flowing, NON-marginal osteophytes of DISH and the bulky asymmetric/non-marginal syndesmophytes of psoriatic/reactive arthritis. Calling any spinal fusion 'bamboo spine/AS' without checking the SI joints is the trap.

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8. Fat tissue

WHAT YOU SEE

Scroll step 2: yellow fat labeled FAT TISSUE

WHAT IT ENCODES

Step 2 is post-inflammatory FATTY METAPLASIA: as active osteitis resolves, the marrow shows fat deposition (bright on T1 MRI), an intermediate repair stage that precedes new bone formation. Recognizing this stage explains how inflammation transitions toward ankylosis and is a marker of prior inflammation on MRI.

BOARD TRAP

Fatty metaplasia (bright on T1) is a REPAIR/intermediate stage, not active inflammation (which is edema bright on STIR/T2) and not yet fused bone — mislabeling the T1-bright fatty change as active disease requiring escalation is the trap.

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9. HLA-B27 scroll

WHAT YOU SEE

Scroll labeled HLA-B27 with 2:1 ratio

WHAT IT ENCODES

HLA-B27 (MHC class I) is positive in ~90% of radiographic AS in some populations and is the strongest genetic risk factor, but most B27-positive individuals NEVER develop disease (low penetrance, ~5%). Radiographic AS shows a male predominance (~2–3:1), while non-radiographic axial SpA is more sex-balanced. B27 is a supportive ASAS classification criterion, not a stand-alone test.

BOARD TRAP

HLA-B27 is SUPPORTIVE, not diagnostic — a positive test in an asymptomatic person does not establish disease, and a negative test does not exclude it. Ordering HLA-B27 as a confirmatory/screening test, or diagnosing AS off a positive B27 alone without imaging and clinical criteria, is the trap.

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10. 20-30 yrs onset

WHAT YOU SEE

Tag reading 20-30 YRS

WHAT IT ENCODES

Axial SpA typically begins in late adolescence to young adulthood (peak ~20–30 years), and symptom onset BEFORE age 45 is a formal ASAS classification criterion. New-onset inflammatory back pain in an older adult is atypical and should prompt consideration of alternatives (malignancy, infection, DISH).

BOARD TRAP

Onset of the back pain after age 45 argues AGAINST axial SpA and should trigger a search for another cause (e.g. metastatic disease, vertebral osteomyelitis, DISH). Diagnosing new axial SpA in a 60-year-old is the trap.

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11. New bony spur

WHAT YOU SEE

Scroll step 3: bone spur labeled NEW BONY SPUR

WHAT IT ENCODES

Step 3 is new bone formation: osteoblast-driven syndesmophytes grow at the vertebral corners and entheses (driven by IL-23/IL-17, BMP, and Wnt signaling with low DKK-1), bridging vertebrae into ankylosis. Crucially, suppressing inflammation does not fully reverse established new bone, underscoring early treatment.

BOARD TRAP

New bone formation/ankylosis is largely IRREVERSIBLE, and anti-inflammatory therapy halts progression rather than dissolving existing syndesmophytes — expecting biologics to reverse a fused bamboo spine is the trap. This is the rationale for treating during the early, edema/reversible stage.

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12. Three scrolls (lung/heart/spine)

WHAT YOU SEE

Three hanging scrolls depicting lung, heart, spinal cord

WHAT IT ENCODES

Extra-articular complications: (1) APICAL pulmonary fibrosis and reduced chest expansion from costovertebral fusion (restrictive pattern); (2) ascending AORTITIS with aortic-root dilation → aortic regurgitation, plus conduction-system disease (AV block); (3) neurologic risk — a rigid, osteoporotic spine fractures easily (even minor trauma) with risk of spinal cord injury and cauda equina syndrome. The 'A' complications: Anterior uveitis, Aortic regurgitation, Apical fibrosis, AV block, Amyloidosis, Achilles enthesitis.

BOARD TRAP

An AS patient with a rigid spine after even MINOR trauma and new neurologic deficits has an unstable spinal FRACTURE until proven otherwise (often at the cervical or thoracolumbar junction) — attributing new back pain/weakness to a routine flare and skipping imaging (CT/MRI) is a dangerous trap. Also, the lung disease is APICAL fibrosis, mimicking TB.

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13. Bonsai with red root

WHAT YOU SEE

Bonsai tree with inflamed red ROOT

WHAT IT ENCODES

Enthesitis — inflammation where tendons/ligaments insert into bone — is a defining, near-pathognomonic feature of the entire spondyloarthritis family, classically at the Achilles tendon and plantar fascia (heel pain). Dactylitis ('sausage digit') is the peripheral counterpart. Enthesitis distinguishes SpA from rheumatoid arthritis, which targets the synovium.

BOARD TRAP

Enthesitis (insertion-site inflammation) is the SpA hallmark, versus the SYNOVITIS of rheumatoid arthritis — picking synovial pannus as the primary lesion of SpA is the trap. Heel/Achilles pain and dactylitis point to spondyloarthritis, not RA.

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14. Bamboo spine / dripping figures

WHAT YOU SEE

Labels BAMBOO SPINE, FLOWING/DRIPPING formation, NORMAL SI joints, woman triangle

WHAT IT ENCODES

The key differential is DISH (diffuse idiopathic skeletal hyperostosis): flowing, 'dripping candle wax' ossification along ≥4 contiguous vertebrae (anterolateral, right-sided in the thoracic spine), with PRESERVED disc spaces and NORMAL, non-fused sacroiliac and apophyseal joints, in OLDER, often obese/diabetic patients. It is non-inflammatory and not HLA-B27-associated.

BOARD TRAP

DISH SPARES the sacroiliac joints and is NON-inflammatory — calling its thick, flowing, non-marginal osteophytes 'ankylosing spondylitis' is the trap. The discriminators: AS has bilateral sacroiliitis + thin marginal syndesmophytes in a young B27+ patient, whereas DISH has normal SI joints + flowing ossification in an older metabolic-syndrome patient.
